HW1: 3D Scene Reconstruction

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1. Implementation

1.1 Reconstruction Pipeline

Each RGB-D frame is unprojected to a point cloud, downsampled, aligned to the previous frame with RANSAC followed by point-to-plane ICP, and the pairwise poses are chained into one map and one camera trajectory.

Major steps

1. **Depth unprojection.** Implemented from scratch with the pinhole model; no Open3D projection function is used. Pixels with finite, positive depth become 3D points. The stored depth is in millimetres and is divided by the depth scale (1000) in `reconstruct()` before unprojection. The captures are 512 × 512 with a 90° horizontal FOV, so fx = fy = W / (2 tan(hfov / 2)) = 256 px and cx = cy = 256 px.

```
Z = D(u,v)/s, \quad X = (u - c_x)\,Z / f_x, \quad Y = (v - c_y)\,Z / f_y
```

2. **Voxelization.** Voxel downsampling, then normals (needed for point-to-plane ICP) and FPFH descriptors for RANSAC.
3. **Global registration.** The current frame is registered to the previous frame with RANSAC on FPFH correspondences.
4. **Local registration.** Open3D point-to-plane ICP refines the RANSAC result.
5. **Scene map and evaluation.** Pairwise transforms are composed (T_global = T_global @ T_cur→prev), the clouds are transformed and accumulated into one map, and the ceiling band is cropped for display. Mean L2 is the mean Euclidean distance between estimated and ground-truth camera positions, with the first camera pose as the reference frame. The estimated trajectory is drawn in red and the ground truth in black.

Settings (voxel size v = 0.05 m unless stated otherwise)

Stage	Parameter	Value
Voxelization	Voxel size	v = 0.05 m (0.04–0.08)
Normals	Search radius, max neighbours	2*v, 30
FPFH	Search radius, max neighbours	5*v, 100
RANSAC	Distance threshold	1.5*v
RANSAC	Points per sample, checkers	3; edge-length (0.9) and distance
RANSAC	Max iterations, confidence	100000, 0.999
ICP	Correspondence threshold	1.5v
ICP	Estimation, max iterations	Point-to-plane, 100

Issues and solutions

- **Double depth scaling.** An early version of the unprojection also divided integer depth by 1000, although `reconstruct()` had already converted to metres. Every cloud shrank 1000×, the map did not render, and Mean L2 was about 4.3 m. The unprojection now takes metres only and just casts to float.
- **Invalid depth.** NaN and infinite depth values are now excluded, and the RGB shape is validated. Didn't not trace any result to it.
- **Sparse FPFH matches.** Open3D prints "too few correspondences after mutual filter" and falls back to the unfiltered correspondences. It appeared around Phase 1 pairs 100–120 and Phase 2 frames 175–250, where voxel yield is low.
- **Reproducibility.** The RANSAC seed is set. On macOS I ran with `OMP_NUM_THREADS=1 KMP_DUPLICATE_LIB_OK=True`. Three seeds on the Phase 2 baseline gave Mean L2 of 0.065, 0.066 and 0.065 m, so sampling noise is about 1 mm there.

1.2 Data Verification

The eight pipeline factors were recorded for both captures. They show where depth and geometric support are thin, but only the gap between ICP and RANSAC fitness separated the pairs that actually failed.

Measurements. All factors are recorded by the first-pass `reconstruct.py` run at v = 0.05 m with the settings in Section 1.1.

Factor	Scope	What it records
RawPointCount	Per image	Valid back-projected points before downsampling
VoxelizedPointYield	Per image	Points left after voxel downsampling
CoarseRANSACFitness	Per consecutive pair	Inlier fraction of the RANSAC result
CoarseRANSACRMSE	Per consecutive pair	RMSE of RANSAC inlier correspondences
CoarseRANSACCorrespondences	Per consecutive pair	Correspondences kept by RANSAC
ICPInlierFitness	Per consecutive pair	Inlier fraction after ICP
ICPInlierRMSE	Per consecutive pair	RMSE of ICP inlier correspondences
ICPInlierCorrespondences	Per consecutive pair	Correspondences kept by ICP

Sources, as given in the assignment: the factor definitions (spec Section 2.2.2 and `ontology/hw1.ttl`), the Open3D global registration tutorial and the course repository.

What the plots show (values read from the factor timelines and distributions)

- **Phase 1 (191 frames).** RawPointCount sits at about 2.62×10^5, which is 512×512, for most frames. It drops around frames 47–55 and strongly between about 70 and 135, to about 1.8×10^5. VoxelizedPointYield spans about 4,000–40,000, with troughs near frames 25, 100–110 and 150. The fitness factors run from about 0.62 to 1.0, with dips at the same places. The RMSE factors are narrow (about 0.01–0.035 m) and do not separate good pairs from bad ones. The correspondence counts follow VoxelizedPointYield.
- **Phase 2 (263 frames).** RawPointCount is full until about frame 195, then fluctuates between about 2.15×10^5 and 2.6×10^5. VoxelizedPointYield spans about 4,000–24,000. Fitness is mostly 0.9–1.0, with dips to about 0.7 near frames 85–90, 105–110, 160–170 and 195–215, and about 0.52 for the final pair (a 78.3 cm step).

Failure cases. Ground truth was used only after the run, to explain which pairs failed. It was not used in any selection query. Errors are per-pair translation errors here are from my own diagnostic script, which matched the official Mean L2 for example at v = 0.05 (2.489 vs 2.4887 m).

Run	Pair	Error	Fitness RANSAC → ICP
Phase 1 baseline, v = 0.05	104←103 (in-place turn)	94.7 cm	0.66 → 0.84
Phase 1 baseline, v = 0.05	105←104 (in-place turn)	48.2 cm	0.73 → 0.88
Old query (154 frames), v = 0.05	106←100 (6-frame gap)	335.2 cm	0.48 → 0.58
Old query (154 frames), v = 0.05	152←142 (150 cm step)	186.9 cm	0.54 → 0.62
Gap query (189 frames), v = 0.05	106←103	127.7 cm, rotation off by 90.3°	0.52 → 0.77
Gap query (189 frames), v = 0.04	103←102	365.0 cm, rotation off by 179.7°	0.47 → 0.54
Gap query (189 frames), v = 0.06	106←103	125.2 cm, rotation off by 0.5°	0.52 → 0.71
All frames (191), v = 0.06	104←103	26.2 cm	0.93 → 0.95
Phase 2 baseline, v = 0.05	225←224 (worst pair)	3.0 cm	0.77 → 0.89 (seed 0)

- **Phase 1 failures are local and compound.** The median pair error is 0.14 cm, so the pipeline is good. A wrong pose is inherited by every later frame. With the 90° error, each later 25 cm step added about 34.5 cm to the accumulated error, as expected from $2 \times 25 \times \sin 45° = 35$ cm.
- **The ugly baseline results cause.** ICP fitness rose by 0.15–0.18 over RANSAC fitness, which suggests ICP had to rescue a wrong coarse alignment. Healthy pairs show almost no change, for example 0.73 to 0.72. Fitness alone is not a safe quality score, because the wrong 90° alignment scored 0.77.

- **Phase 2 has no failed pairs.** The worst pair is off by 3.0 cm. The largest jumps in accumulated error are 1.9 cm (225←224) and 1.8 cm (220←219), both in the frames where RawPointCount drops.

Adjustments made from this evidence

- **Old query.** Dropped frames with VoxelizedPointYield < 13200 or ICPInlierFitness < 0.7. It removed whole runs of consecutive frames, so surviving neighbours were up to 188 cm and 90° apart. Mean L2 rose from 3.1741 to 12.9199 m.
- **Gap query.** Drops a frame when ICP fitness exceeds RANSAC fitness by more than 0.1 (FILTER (?iv - ?rv > 0.1), where ic is ICP value, rv is RASNAC value). The 0.1 threshold comes from two failing pairs, so it is a hypothesis, not a validated rule. It removed frames 104 and 105 (189 frames kept) and gave 2.4887 m, but it created a new pair 106←103 that the recorded factors never measured.
- **Voxel size.** Swept 0.03–0.08 (Section 2.3). The same stretch around frames 103–106 failed at every size, in different ways.
- **Phase 2.** The gap query returned all 263 frames, so the selected run equals the baseline. Pair 225←224 had a RANSAC fitness of 0.77 with seed 0 but 0.89 and 0.86 with seeds 1 and 2, so the rule's signal is seed-sensitive.

```
PREFIX hw1: <http://taica.course/hw1/ontology#>
PREFIX exp: <http://taica.course/hw1/experiment/>
SELECT DISTINCT ?frameIndex WHERE {
    exp:p1_baseline hw1:onBatch ?batch .
    ?batch hw1:hasFrame ?frame .
    ?frame hw1:frameIndex ?frameIndex ;
           hw1:hasDepthImage ?depth .
    FILTER NOT EXISTS {
        ?r a hw1:DepthPairFactor ;
           hw1:inExperiment exp:p1_baseline ;
           hw1:hasCurrentFrame ?depth ;
           hw1:value ?rv ;
           hw1:hasDefinition hw1:CoarseRANSACFitness .
        ?i a hw1:DepthPairFactor ;
           hw1:inExperiment exp:p1_baseline ;
           hw1:hasCurrentFrame ?depth ;
           hw1:value ?iv ;
           hw1:hasDefinition hw1:ICPInlierFitness .
```

```
        FILTER (?iv - ?rv > 0.1)
    }
}
ORDER BY ?frameIndex
```

2. Results and Discussion

2.1 Visual Results

All maps are shown with the ceiling removed, the estimated trajectory in red and the ground-truth trajectory in black.

Phase 1: apartment_0, first floor



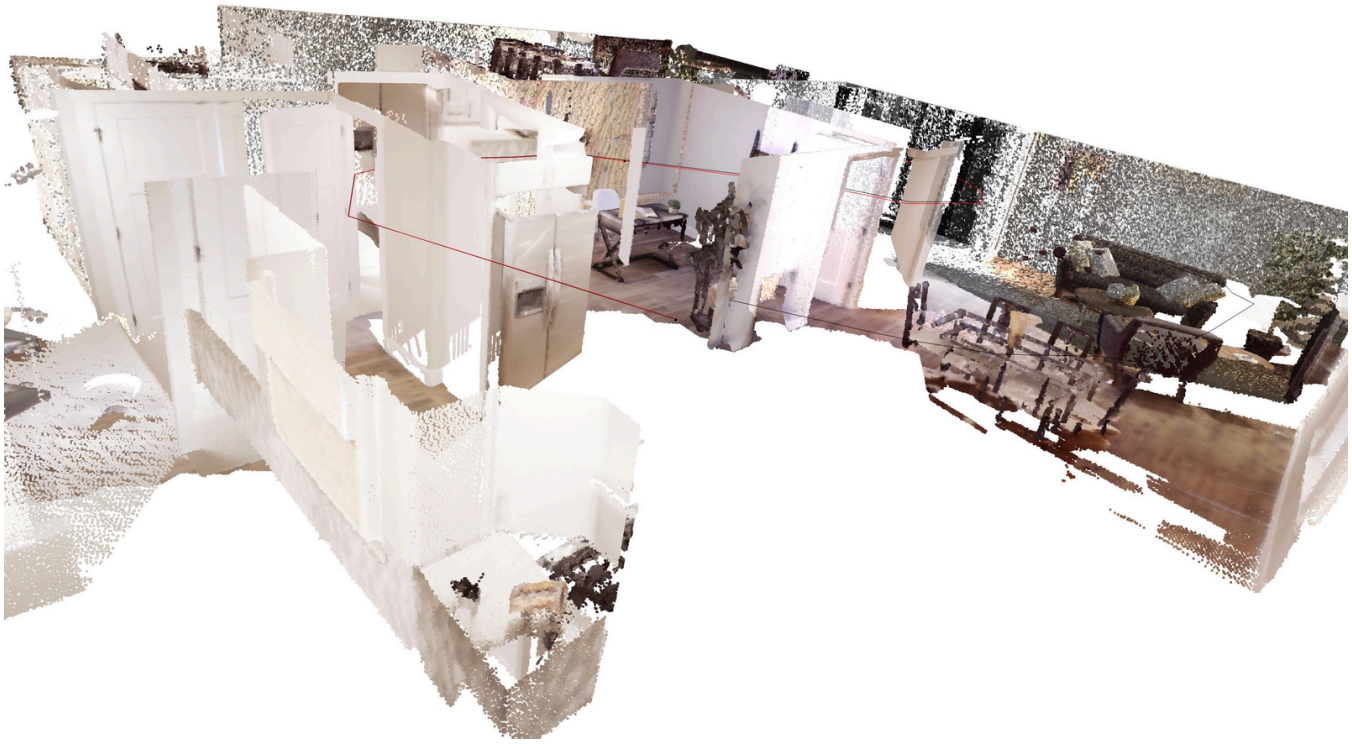


Figure 1. Baseline (191 frames, v = 0.05, Mean L2 = 3.1741 m), top view and oblique view.

Figure 2. Old query (154 frames, v = 0.05, Mean L2 = 12.9199 m). No reconstruction image is shown (see the note to the TA below).

Note to the TA: I forgot to record the reconstruction screenshots and the run times for the old-query run (12.9199 m), the gap-query run (2.4887 m) and the gap-query preview at v = 0.04 (2.874 m). Their Mean L2 values were copied from my terminal output at the time, so no figure or run time is given for them.

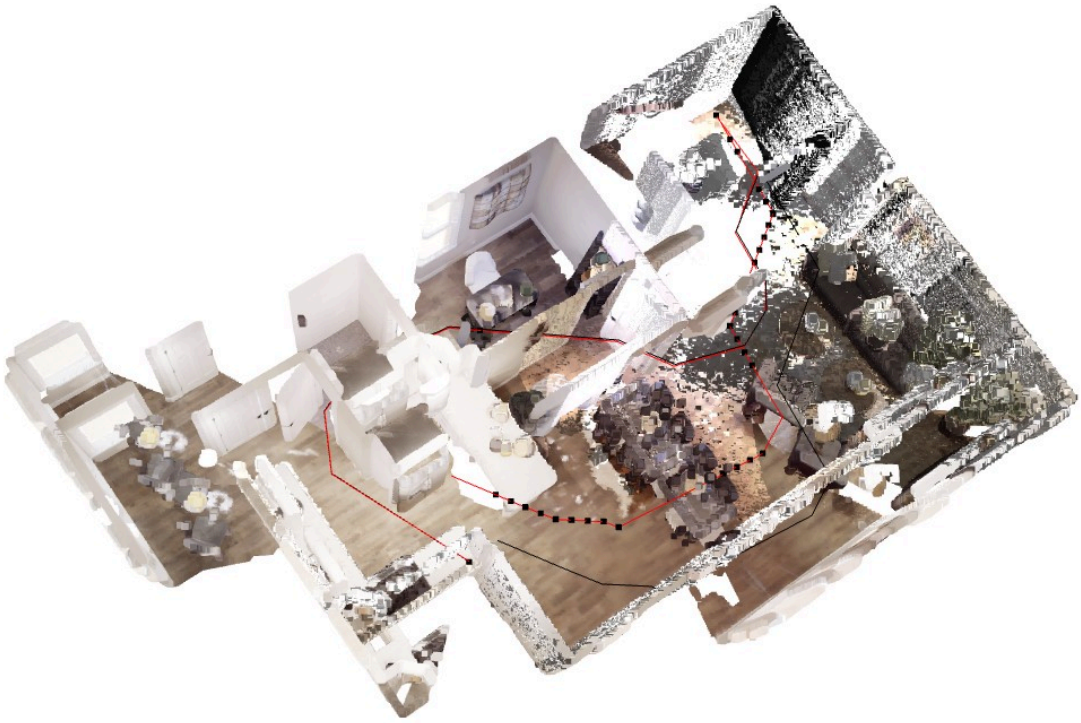


Figure 3. Final Phase 1 configuration (191 frames, v = 0.06, Mean L2 = 0.1104 m).

Phase 2: apartment_0, second floor

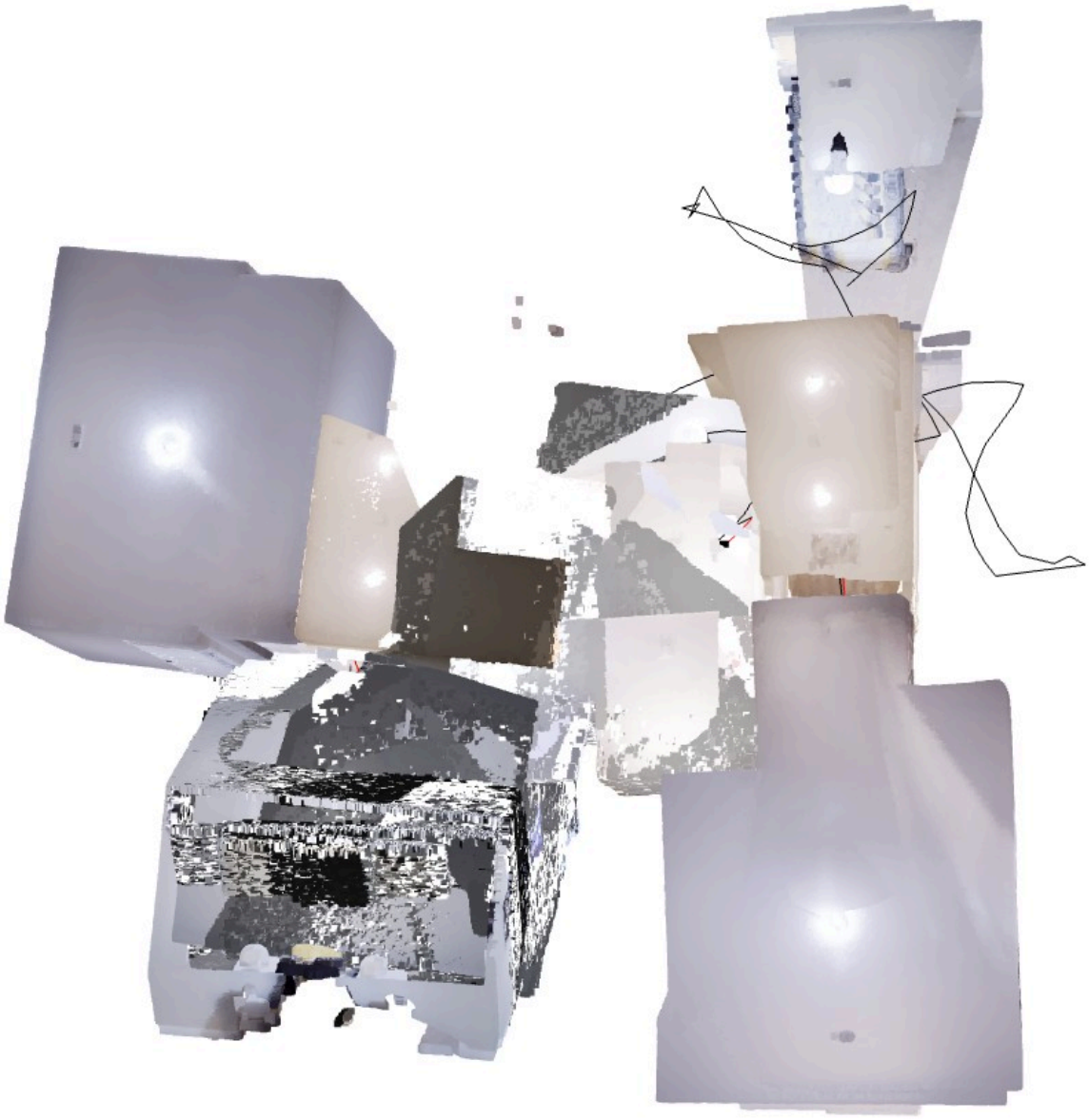


Figure 4. Earlier Phase 2 collection with poor data collection (403 frames, Mean L2 about 5.4391 m). The rooms are not assembled into a coherent floor plan.

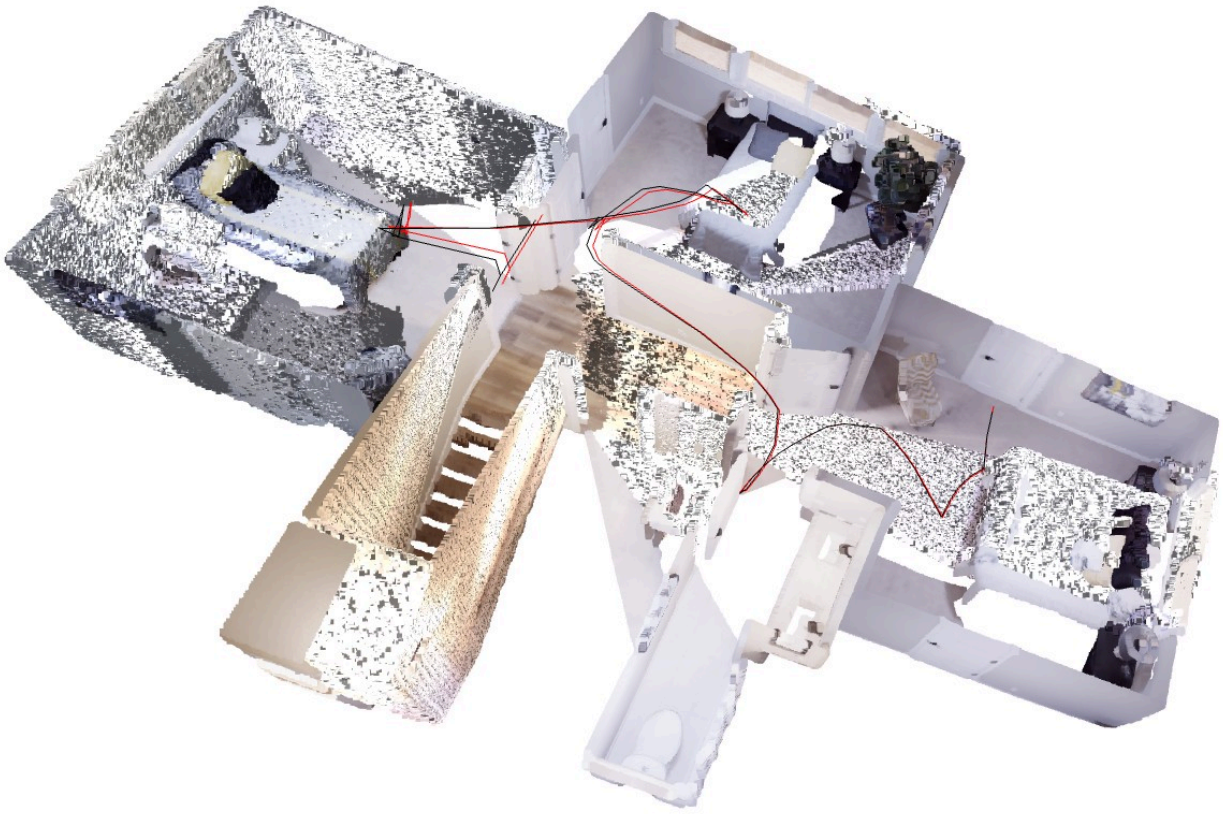


Figure 5. Baseline (263 frames, v = 0.05, Mean L2 = 0.0646 m).

Bonus: custom point-to-plane ICP



Figure 6. Custom ICP, Phase 1 (191 frames, v = 0.06, Mean L2 = 0.1098 m).

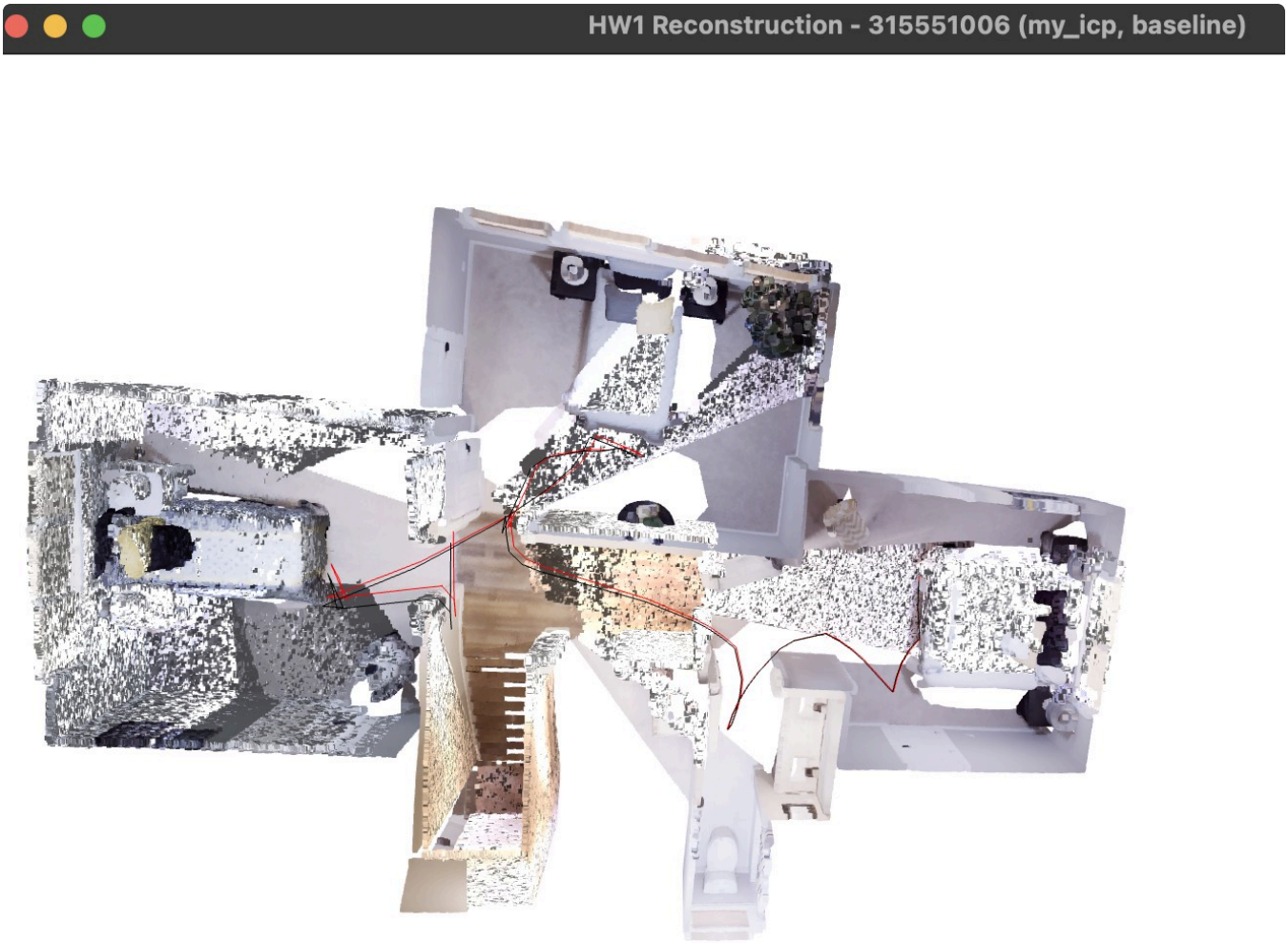


Figure 7. Custom ICP, Phase 2 (263 frames, v = 0.05, Mean L2 = 0.0650 m).

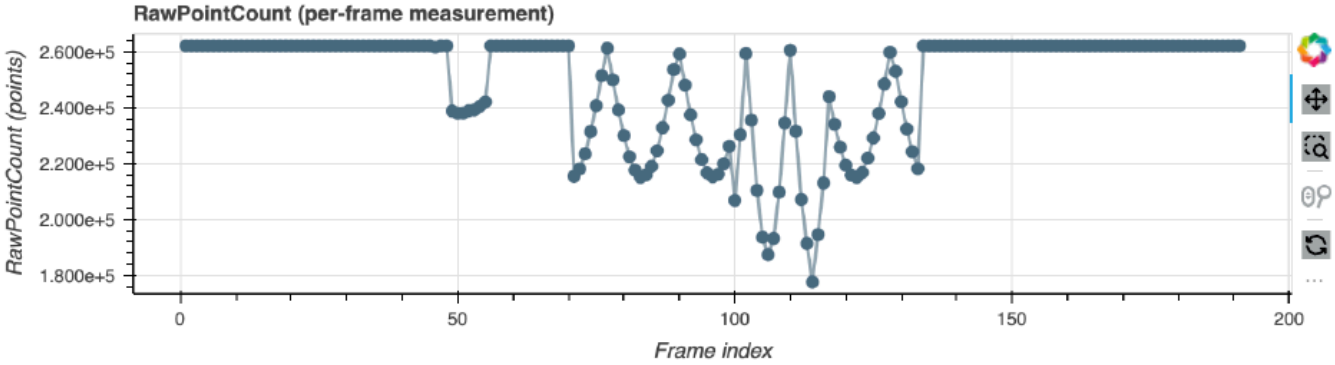
Verification plots

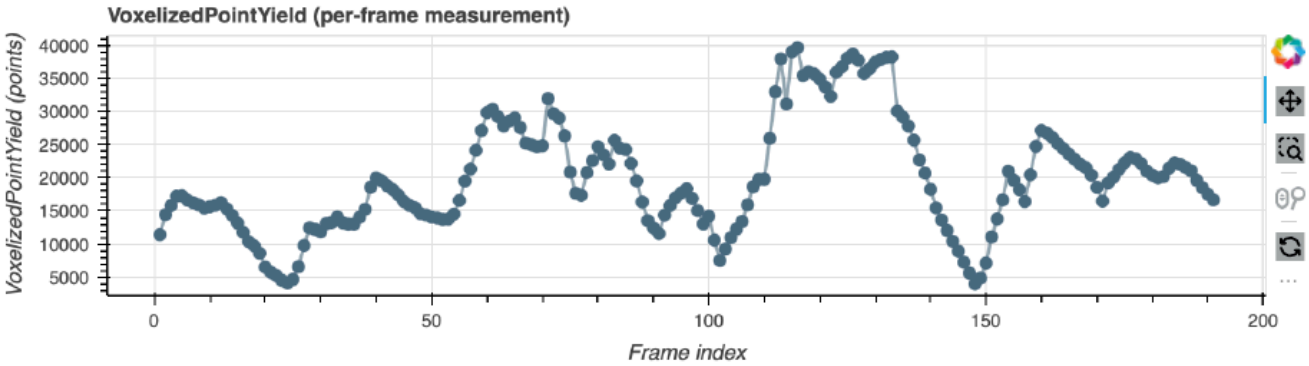
Factor timelines | Factor distributions | Selection tradeoff

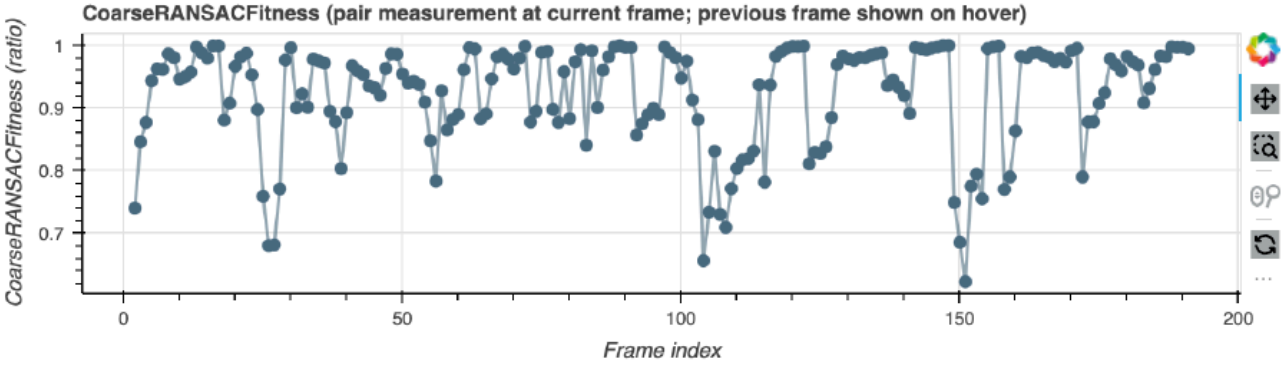
Experiment p1_baseline

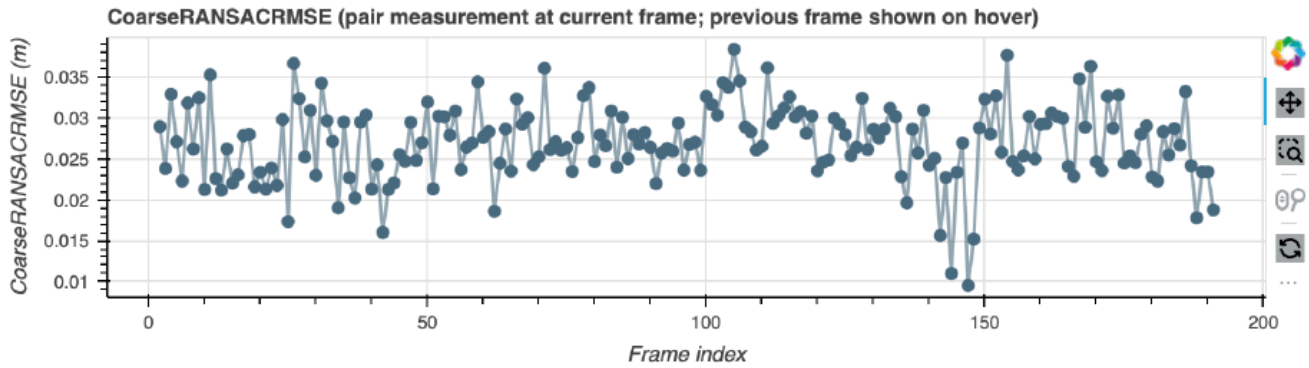
Batch contains 191 frames. The tradeoff plot uses only recorded baseline/selected runs. The schema stores one run per mode, so one experiment has at most two points; no reconstruction sweep is run or inferred.

Tip: Hover over any point in the factor timelines to inspect the corresponding RGB and depth image pair.











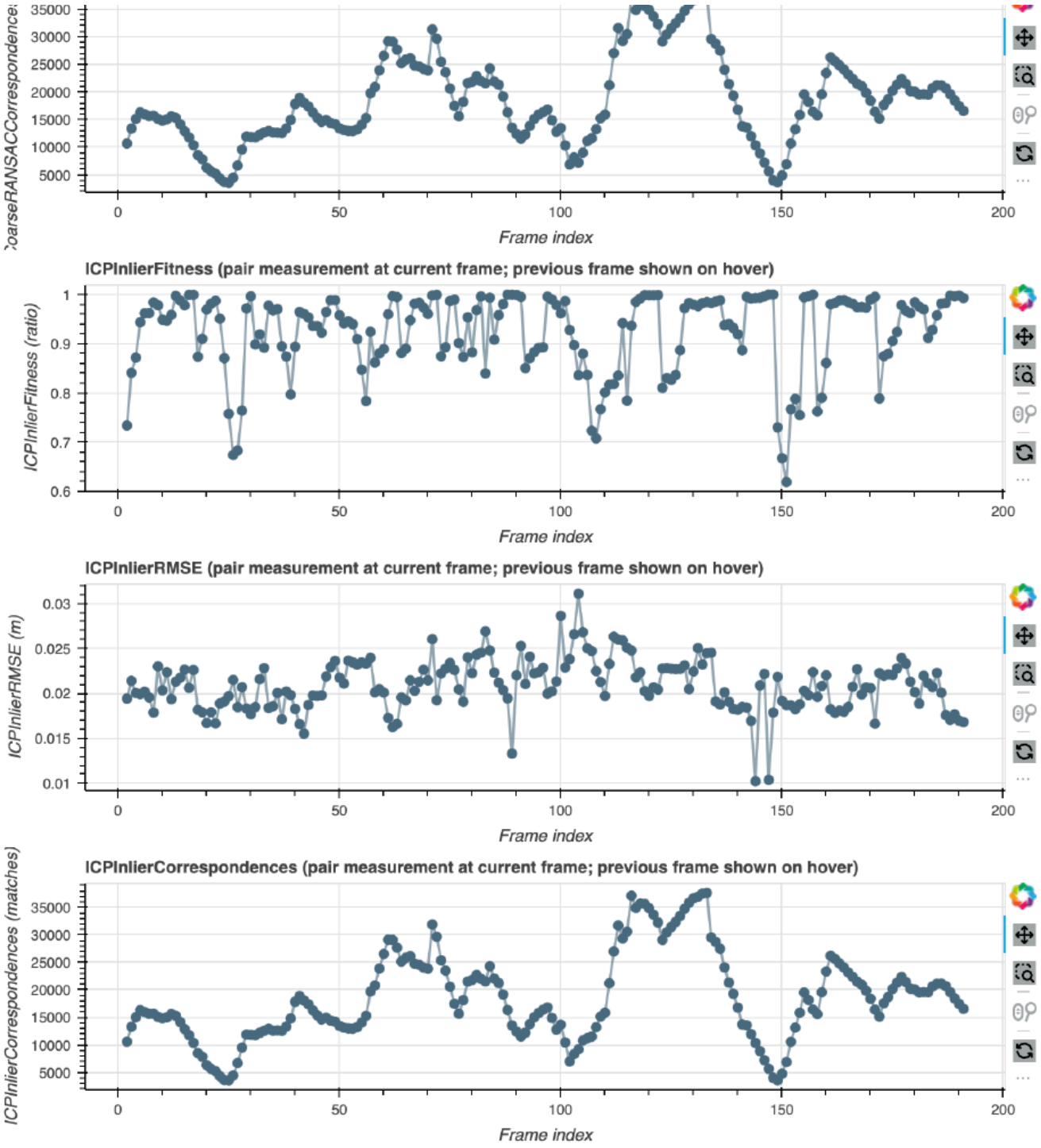


Figure 8. Phase 1 factor timelines (experiment `p1_baseline`).

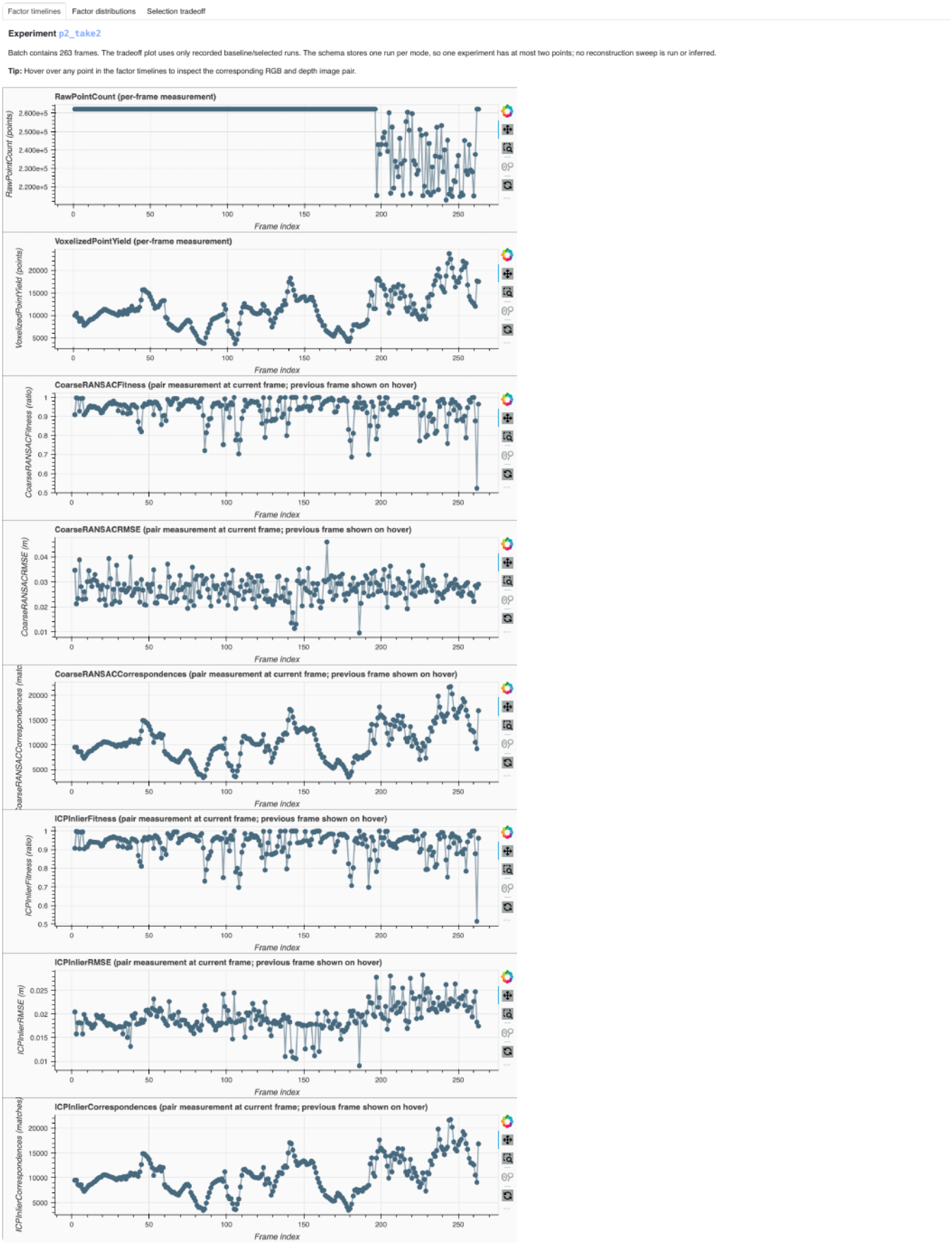


Figure 9. Phase 2 factor timelines (experiment `p2_take2`).

Factor timelines | Factor distributions | Selection tradeoff

Experiment `p2_baseline`

Batch contains 403 frames. The tradeoff plot uses only recorded baseline/selected runs. The schema stores one run per mode, so one experiment has at most two points; no reconstruction sweep is run or inferred.

Tip: Hover over any point in the factor timelines to inspect the corresponding RGB and depth image pair.

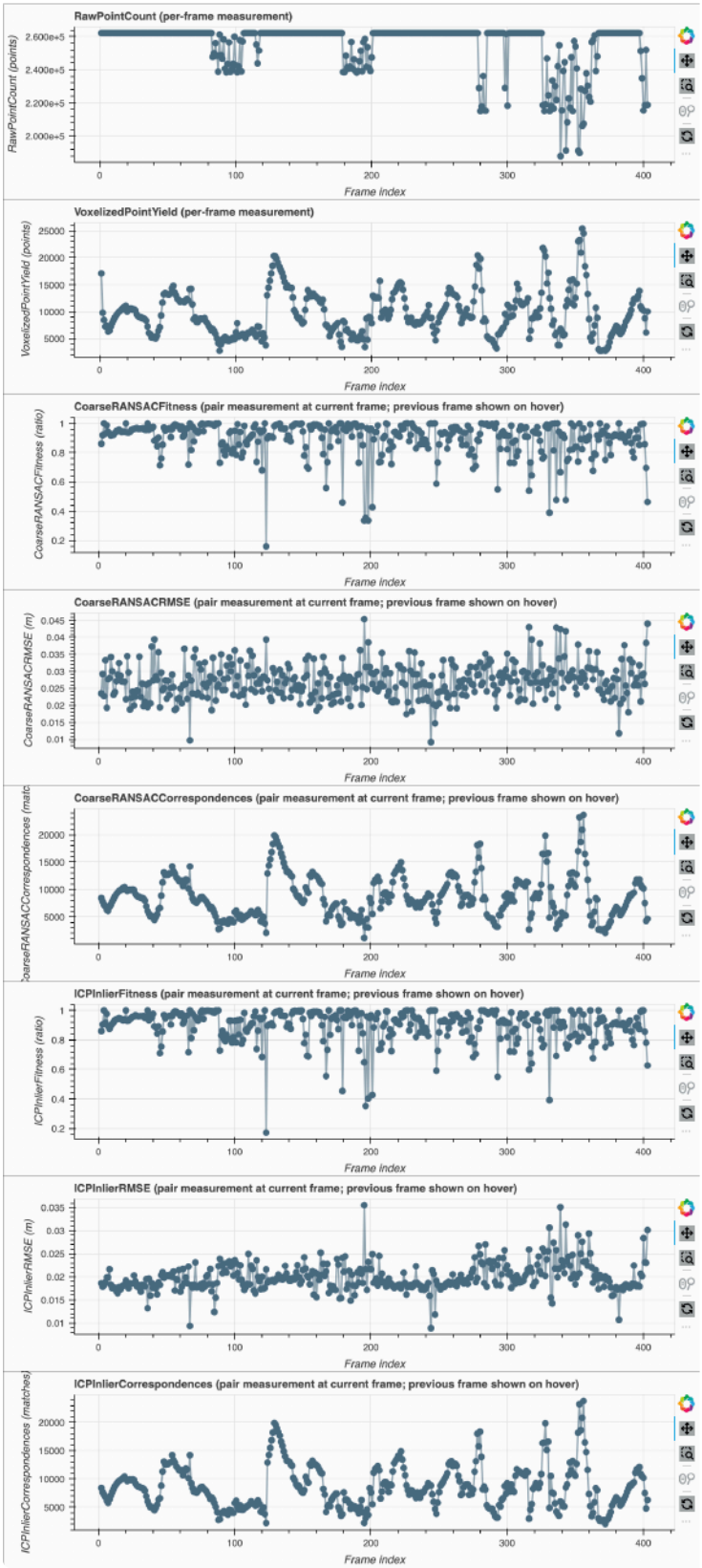


Figure 10. Factor timelines for the 403-frame Phase 2 collection (experiment `p2_baseline`). CoarseRANSACFitness and ICPInlierFitness drop to about 0.17 near frame 120 and to about 0.35 near frames 195–200.

2.2 Comparison Table

Phase 2 reaches 0.0646 m with the unmodified pipeline. Phase 1 needed intervention: the official result improved from 3.1741 m at v = 0.05 to 0.1104 m at v = 0.06 with all frames, while frame selection alone (the gap query) reached 2.4887 m.

Phase	Configuration	Frames	Mean L2 (m)	Total time (s)	Hyperparameters	Source
1	Baseline	191	3.1741	558.24	v = 0.05, all frames	Official
1	Old query	154	12.9199	not recorded	v = 0.05; drop VoxelizedPointYield < 13200 or ICPInlierFitness < 0.7	Official
1	Gap query	189	2.4887	not recorded	v = 0.05; drop frames where ICP fitness − RANSAC fitness > 0.1	Official
1	Gap query	189	2.874	not recorded	v = 0.04	Preview
1	Gap query	189	0.580	447.12	v = 0.06, seed 0	Preview
1	Gap query	189	> 4		v = 0.03 and 0.08 (individual values not recorded)	Preview
1	All frames	191	0.211		v = 0.06, seed 2	Preview
1	All frames	191	0.1104	333.08	v = 0.06	Official
2	Baseline	263	0.0646	374.53	v = 0.05	Official
2	Baseline, seeds 0 / 1 / 2	263	0.065 / 0.066 / 0.065	not recorded	v = 0.05	Preview
2	Baseline	263	0.075	545.98s	v = 0.04	Preview
2	Baseline	263	0.092	175.72s	v = 0.06	Preview
2	Baseline	263	0.160	not recorded	v = 0.08	Preview
2	Earlier collection (poor data collection)	403	about 5.4391	not recorded	settings not recorded	Terminal output

Official values are printed by `reconstruct.py`. Previews are chained by our diagnostic script on the same frames. It reproduced the official values at v = 0.05 (2.489 vs 2.4887 m in Phase 1, 0.065 vs 0.0646 m in Phase 2), but a different voxel size changes the recorded factors, so previews at other sizes are not official runs. A run time is given only where I recorded it; “not recorded” marks runs I forgot to time (see the note to the TA in Section 2.1).

2.3 Discussion and Findings

The pipeline registers almost every pair accurately in both phases. Mean L2 is decided by the few pairs that fail, because each pose is chained into every later frame.

Voxel size. Mean L2 of the chained trajectory in metres. Phase 1 uses the 189-frame gap-query selection. Blank cells were not run.

Voxel (m)	Phase 1 RANSAC + ICP	Phase 1 ICP only	Phase 2 RANSAC + ICP	Phase 2 ICP only
0.03	> 4			
0.04	2.874	2.031	0.075	1.987
0.05	2.489	1.675	0.065	1.701
0.06	0.580	1.363	0.092	1.540
0.08	> 4		0.160	0.738

Phase comparison

- **The tail decides the result.** The median pair error is 0.14 cm in both phases. Phase 1 had pairs off by 48–335 cm. The worst Phase 2 pair is off by 3.0 cm, so Phase 2 reaches 0.0646 m with no intervention.
- **Collection differs.** Phase 2 has a median step of 0 cm (mostly in-place turns), a maximum step of 78.3 cm and a maximum rotation of 18° between consecutive frames. In the Phase 1 baseline the largest step is 25.3 cm and the largest rotation is 10°.
- **Failures needed several conditions at once.** In Phase 1, missing depth (RawPointCount down to about 1.8×10^5 between frames 70 and 135), low voxel yield and fitness dips coincide around frames 100–110, where the failing pairs are. In Phase

2, depth loss starts only after frame 195 and is milder (about 2.15×10^5 at the lowest), and no pair failed. This is consistent with the data but not a controlled test.

- **RANSAC matters in Phase 2.** ICP from identity gives 1.701 m against 0.065 m with RANSAC.

Reconstruction adjustments and quality improvements

- **Frame selection can hurt.** The old query removed whole runs of consecutive frames and raised Mean L2 from 3.1741 to 12.9199 m, because neighbours that were 25 cm apart became up to 188 cm and 90° apart.
- **A narrower rule helped, with a side effect.** The gap query dropped only frames 104 and 105 and gave 2.4887 m. It created a pair (106←103, a 3-frame turn) that the recorded factors never measured, and that pair failed with a 90.3° rotation error.
- **Voxel size is not monotonic.** In Phase 1 the same stretch failed at every size, in different ways. The best preview, 0.580 m, was at v = 0.06. Because poses are chained, one bad pair dominates the result.
- **All frames at v = 0.06.** The official run gave 0.1104 m in 333.08 s, against 558.24 s at v = 0.05, so a larger voxel alone removed most of the Phase 1 error without dropping any frame. The diagnostic preview with seed 2 gave 0.211 m (worst pair 26.2 cm, rotation errors below 0.5°), so at this voxel size the result may still depend on the RANSAC seed, which we did not vary in the official run. The 0.580 m row differs in both frame set and seed, so it is not a controlled comparison.
- **Phase 2 selection.** The gap query dropped nothing, so that run equals the baseline. In Phase 2, v = 0.05 had the lowest Mean L2 of the sizes tested, but the 0.04 and 0.05 results differ by only 1 cm.

3. Questions

1. What happens if ICP is performed without global registration (RANSAC)? Why?

ICP only converges to the nearest local minimum, so without a RANSAC start it works when consecutive frames are already close and fails when they are far apart. We ran ICP from the identity transform on every pair next to the normal RANSAC + ICP chain:

- **Small motions.** The median per-pair error is the same with and without RANSAC (0.14 cm in both phases), because consecutive frames are close.

- **Phase 2.** ICP from identity gave a chained Mean L2 of 1.701 m against 0.065 m with RANSAC. The worst pair error was 139.2 cm without RANSAC and 3.0 cm with it.
- **Large gaps.** In the old-query run, pair 32←14 (188 cm apart) had a 318.7 cm error without RANSAC and 72.6 cm with it.
- **RANSAC can also fail.** In weak-feature stretches of Phase 1 the identity start was better on several pairs: 106←103 (7.6 vs 127.7 cm), 106←100 (27.0 vs 335.2 cm) and 104←103 (26.6 vs 94.7 cm). On the 189-frame selection, ICP-only gave a lower chained Mean L2 at v = 0.05 (1.675 vs 2.489 m) but a higher one at v = 0.06 (1.363 vs 0.580 m). Choosing the better of the two by ICP fitness gave no change at v = 0.05 or 0.06, because the wrong alignments had high fitness. It did lower the chain to 1.233 m at v = 0.04.

2. Describe the choices used to improve ICP stability, such as voxel size, correspondence thresholds, or robust losses.

- **Point-to-plane metric.** All ICP runs use point-to-plane with normals estimated at 2v (at most 30 neighbours), as the assignment requires. We did not compare it with point-to-point on the real captures.
- **Correspondence distance thresholding.** The ICP and RANSAC thresholds are both 1.5v, so changing the voxel size changes every stage at once. In Phase 2, ICP-only improved as the voxel grew (1.987 m at 0.04 to 0.738 m at 0.08), which fits a wider capture range. RANSAC + ICP did not follow (0.065 m at 0.05, 0.160 m at 0.08).
- **Voxel size.** Swept from 0.03 to 0.08. The Phase 1 result was not monotonic and Phase 2 was best at 0.05 (Section 2.3).
- **Convergence and reproducibility.** Open3D ICP is capped at 100 iterations. RANSAC uses up to 100000 iterations at 0.999 confidence with a fixed seed.
- **Frame selection.** Selection rules were chosen to avoid creating long gaps between kept frames, after the old query raised Mean L2 to 12.9199 m.
- **Not used.** Robust loss kernels and multi-scale downsampling were not tested.

3. How do the observed data-quality factors and collection choices relate to reconstruction performance and Mean L2 Distance?

- **Missing depth and low yield mark where to look.** In Phase 1, the drop in RawPointCount, the troughs in VoxelizedPointYield and the fitness dips coincide around frames 100–110, where the failing pairs are. In Phase 2, RawPointCount drops

after frame 195 and the largest accumulated-error jumps (pairs 220←219 and 225←224, 1.8–1.9 cm) are in that segment.

- **A dip does not always mean failure.** Phase 1 fitness also dips near frames 25 and 150, but those pairs have errors of about 1 cm or less. The RMSE factors barely vary, and the correspondence counts mirror VoxelizedPointYield.
- **The ICP-over-RANSAC fitness gap separated the failures.** The two failing baseline pairs rose by 0.15–0.18 between the RANSAC and ICP stages, while healthy pairs hardly changed.
- **Collection choices.** Small steps keep ICP in its basin. The old query removed frames and increased gaps, which raised Mean L2 from 3.1741 to 12.9199 m. Phase 2′s capture (mostly in-place turns, steps of 20 cm) produced no failed pair.
- **Chaining links the two.** Mean L2 is dominated by the worst pair. A 90.3° rotation error on one pair added about 34.5 cm per later 25 cm step and gave 2.4887 m. Phase 2′s worst pair of 3.0 cm gave 0.0646 m.

4. (Optional) Bonus Track: Custom ICP Implementation

The custom point-to-plane ICP matches Open3D′s on both phases: 0.0650 against 0.0646 m in Phase 2 and 0.1098 against 0.1104 m in Phase 1, gaps of 0.4 and 0.6 mm. The Phase 2 gap is below the roughly 1 mm seed noise measured there. Total run time was 1.0% and 1.8% longer (378.27 vs 374.53 s and 339.04 vs 333.08 s), from one run each. It is implemented in `my_local_icp_algorithm` and does not replace the required Open3D pipeline. The two custom-ICP reconstructions are Figures 6 and 7 in Section 2.1.

Implementation approach

- **Correspondences.** A `cKDTree` is built once on the target cloud per pair, and the transformed source points are matched to their nearest neighbours each iteration. Pairs farther apart than 1.5v are rejected (the same threshold as the Open3D ICP), and the loop stops if fewer than 10 pairs remain.
- **Update step.** Point-to-plane, using the target normals from the preprocessing step (estimated on a copy if missing). For a small rotation w and translation t, each pair gives one equation of a 6-unknown linear system, solved by least squares:

```
A_i = [\, p_i \times n_i \;\; n_i \,], \qquad b_i = n_i \cdot (q_i - p_i),
\qquad A x = b, \quad x = [w;\, t]
```

- **Rigid update.** The small-angle vector is converted back to a rotation with Rodrigues' formula and composed into the running 4 × 4 transform (T = ΔT @ T), so the result stays a valid rotation.
- **Convergence.** At most 60 iterations, stopping when the mean inlier error changes by less than 1e-6.
- **Reported metrics.** Fitness, inlier RMSE and the correspondence count are evaluated once at the final transform, so they describe the returned pose. These values feed the ICPInlier* factors.

Comparison

Phase	ICP	Frames	Mean L2 (m)	Total time (s)	Hyperparameters
1	Custom (`p1_bonus` _custom)	191	0.109832	339.04	v = 0.06
1	Open3D, same settings	191	0.110422	333.08	v = 0.06
2	Custom (`p2_bonus` _custom)	263	0.064963	378.27	v = 0.05
2	Open3D (`p2_take2`)	263	0.064608	374.53	v = 0.05

On synthetic noisy clouds with partial overlap, the custom ICP and Open3D's point-to-plane ICP returned the same transform error (0.03° rotation, 0.5 mm translation), the same fitness (0.882) and the same number of correspondences (11,086). The custom output stayed a valid rotation.

Performance enhancements

- **Correspondence search.** One KD-tree per pair instead of a brute-force search.
- **Threshold selection.** 1.5v, matching the Open3D ICP so the two backends are compared on the same footing.
- **Convergence criteria.** An iteration cap of 60 and a mean-error tolerance of 1e-6.
- **Degenerate geometry.** `lstsq` returns a minimum-norm update when the view is nearly planar, instead of raising an error.
- The correspondence list is built once at the end, not on every iteration.
- **Robust weighting.** Not used.
- **Runtime-accuracy tradeoff.** The custom ICP is written in NumPy and Open3D's is compiled C++, so the custom version is expected to be slower. The measured timings are in the comparison table.